PERGAMON

Radiation Physics and Chemistry 59 (2000) 255–275

Radiation Physics and Chemistry

www.elsevier.com/locate/radphyschem

Inelastic-collision cross sections of liquid water for interactions of energetic protons ☆

Michael Dingfelder*, Mitio Inokuti [1], Herwig G. Paretzke

GSF – National Research Center for Environment and Health, Institute of Radiation Protection, Ingolstädter Landstraße 1, D-85764 Neuherberg, Germany

Received 31 August 1999; accepted 28 January 2000

Abstract

Cross-section data for inelastic interactions of energetic protons with liquid water, for use, e.g. as input in track structure analysis, are derived for an energy range from 0.1 keV to 10 GeV. At proton kinetic energies above about 500 keV, the first Born approximation and the dielectric-response function determined earlier are used. At proton energies above several hundred MeV in particular, the Fermi-density effect is also incorporated. At energies below about 500 keV, which corresponds to a residual range of about 8.9×10^{-6} m, cross-section values are derived semi-empirically by an extensive and critical analysis of experimental and theoretical information concerning not only cross sections for individual processes such as ionisation, excitation, and charge transfer but also stopping power and other relevant quantities. Spectra of secondary electrons resulting from ionising collisions are also presented. The analysis also includes considerations of phase effects on cross sections. © 2000 Elsevier Science Ltd. All rights reserved.

1. Introduction

Track structure calculations of charged particles (Paretzke, 1987; Ritchie et al., 1991) are useful for the understanding of early physical and chemical stages of radiation actions on matter in general. They are used especially in research on the effects of radiation on the biological cell or on DNA (Friedland et al., 1999),

☆ Soon after acceptance of this manuscript we became aware of a work on a similar topic entitled "Calculations of electronic stopping cross sections for low-energy protons in water" by S. Uehara, L.H. Toburen, W.E. Wilson, D.T. Goodhead and H. Nikjoo, also accepted for publication in Radiation Physics and Chemistry (this volume). The authors calculated the electronic stopping cross section for low-energy protons in water vapour using the charge state approach and taking into account excitation and ionisation by protons and hydrogen atoms as well as charge changing processes. Whenever possible pure numerical fits of experimental data were used to represent the interaction cross sections for certain processes, otherwise use of published semi-empirical models for water vapour were made. In this way the calculated stopping cross section overestimates tabulated and experimental values within about 20% for particle energies below 300 keV. On the other hand, this work concentrates on liquid water and uses also the tabulated and experimental stopping cross sections of water in different phases (vapour, liquid and ice) to include phase effects and to adjust semi-empirical models for proton impact in liquid water. In this sense the two papers are complementary and agree in common results.

* Corresponding author. Tel.: +49-89-3187-4113; fax: +49-89-3187-3363.

E-mail address: dingfelder@gsf.de (M. Dingfelder).

[1] Permanent address: Argonne National Laboratory, Physics Division, Argonne, IL 60439, USA.

0969-806X/00/$ - see front matter © 2000 Elsevier Science Ltd. All rights reserved.
PII: S0969-806X(00)00263-2

where

$$\Lambda(E,\beta^2)=\ln\left(\frac{2mv^2Q_1}{I^2}\right)-\ln(1-\beta^2)-\beta^2, \tag{32}$$

and Q_1 is the upper limit of Q, which is set to $Q_1 = 1$ Ry for evaluation. As we expect, the resulting values of $\Lambda(E,\beta^2)$ are insensitive to Q_1. In other words, the use of $\Lambda(E,\beta^2)$ alone leads to the Bethe stopping power, and the presence of $\Delta(E,\beta^2)$ leads to S including the Fermi-density effect.

We interpret Eq. (31) as follows. The Fermi-density effect reduces the probability of energy transfer E from $\Lambda(E,\beta^2)\eta_2(E,0)$ to $[\Lambda(E,\beta^2)-\Delta(E,\beta^2)]\eta_2(E,0)$, apart from the factor in front of the integral of Eq. (31). In other words, the cross section $d\Sigma/dE$ is modified by the factor

$$\phi(E,\beta^2)=1-\frac{\Delta(E,\beta^2)}{\Lambda(E,\beta^2)}. \tag{33}$$

Fig. 4 shows $\phi(E,\beta^2)$ for liquid water for several fixed incident proton energies ($0.55\le\beta^2\le0.95$) as a function of energy transfer E. ϕ is a monotonously increasing function starting with values below unity for small energy transfers and reaching a constant value slightly above unity for high energy transfers. For lower proton energies ($\beta^2 = 0.55$) the factor is almost constant and close to unity, $\phi\approx1$, for all energy transfers. The Fermi-density effect does not influence the cross section visibly. With increasing proton energies ($\beta^2 > 0.55$) ϕ decreases for smaller energy transfers while the constant value for higher energy transfers slightly increases. Now, the Fermi-density effect modifies the differential cross section significantly and reduces the total cross section for higher velocities β^2.

2.4. The applicability of the first born approximation (FBA)

An important difference between an electron and a proton (or a heavier charged particle) concerns the justification of the FBA. The universal criterion, applicable to all particles, is that the speed v of an incident particle far exceeds the mean orbital speed v_m of a molecular electron that receives energy. Then, the (doubly) differential cross section, Eq. (1), is applicable (except at extremely high values of the momentum transfer, at which the cross section is very small). However, the cross section differential in energy transfer, Eq. (2), can be derived also from the first-order treatment in the semi-classical approximation as explained by Bethe and Jackiw (1997). This derivation rests on the condition that the momentum Mv of the incident particle far exceeds the mean orbital momentum mv_m of the molecular electron, which holds down to much lower speeds for a proton or a heavier particle. Furthermore, the electron-exchange effect, as discussed in Section 3.4 of Dingfelder et al. (1998), affects the ionisation cross section by electron collisions at moderately high speeds.

At lower speeds, effects not included in the FBA need to be accounted for. They include effects of the distortion of incident waves, and coupled-channel effects. Inspection of coupled-channel calculations on proton collisions with hydrogen, helium, and other cases (Kimura and Lane, 1990; Schiwietz, 1990; Grande and Schiwietz, 1992) indicate how these effects can influence cross sections, although we are aware of no similar calculations specifically on liquid water. A lower limit of the validity of the FBA might be 200 keV, and more conservatively in the range from 300 to 500 keV. At lower energies we use a semi-empirical approach fully described in Section 3.

2.5. Results

Using the semi-empirical determination (Dingfelder et al., 1998) of the dielectric-response function $\epsilon(E,K)$, as described in Section 2.2, we calculated the differential cross sections using Eq. (2). The integration over momentum-transfer needs a careful treatment, because of the sharply peaked energy-loss function. Furthermore, the main contributions to the integrand of Eq. (2) are often concentrated in a narrow interval around the Bethe ridge. Thus we separated the integral in several integrals using linear and logarithmic integration schemes based on a Romberg method (Press et al., 1992) with a trapezoidal formula. For relativistic energies ($\tau > 10$ MeV) we used the relativistic Bethe formula (Eq. (9)). The determination of the Bethe coefficients $A(E)$ and $B(E)$ is discussed in detail by Dingfelder et al. (1998).

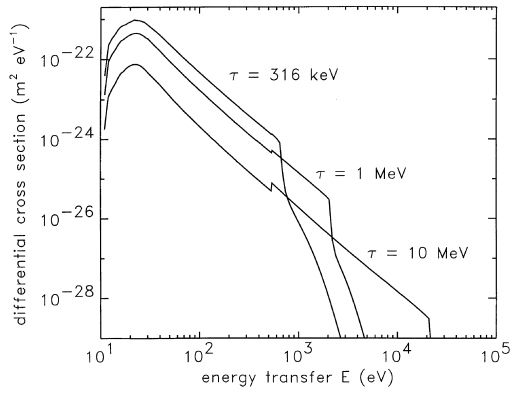

Fig. 5. Energy differential cross sections $d\sigma/dE$, summed over all ionisation shells, vs. energy transfer E for different incident proton energies τ (316 keV, 1 and 10 MeV).

Fig. 5 shows the resulting energy differential cross sections, $d\sigma/dE = N^{-1}d\Sigma/dE$, summed over all five molecular ionisation shells of water for different incident proton energies (from top to bottom $\tau = 316$ keV, 1 and 10 MeV) versus the energy transfer E. The presence of the K-shell is clearly seen in the lower curves, while it is hidden in the sharp decrease of the cross section in the upper curve. This decrease of the cross section is related to the maximum energy transfer from the proton to a free electron, which is (non-relativistically) four times the kinetic energy of an electron of the same speed, $E_{max} = 4(m/M)\tau$. The shown range of incident proton energies corresponds to the range of validity of the non-relativistic FBA.

The total ionisation cross section $\sigma = \int(d\sigma/dE)\,dE$ is shown in Fig. 6 for proton kinetic energies from 1 keV to 10 GeV. It has been calculated in three parts: (1) the semi-empirical approach below 500 keV, (2) the FBA between 500 keV and 10 MeV, and (3) the relativistic Bethe formula above 10 MeV. The Fermi-density correction is not included in the data shown in this figure. The dotted curve shows the failure of the FBA for incident proton energies below 500 keV and for relativistic energies above 10 MeV. For comparison, the electron-impact ionisation cross section, calculated earlier (Dingfelder et al., 1998), is plotted against the energies of protons at the same speed (dashed curve). For electrons the FBA fails at speeds much higher than for protons. One sees also the very weak mass dependence of the Born formula. The small differences between the dashed curve and the solid curve at higher speeds arise from the present treatment of the K-shell different from the calculation for electrons (Dingfelder et al., 1998). The electron and proton cross sections should agree at the same speed in the Bethe-formula when the same dielectric-response function is used; mass-dependent terms only occur in higher order correction terms, which are not included in Eq. (9).

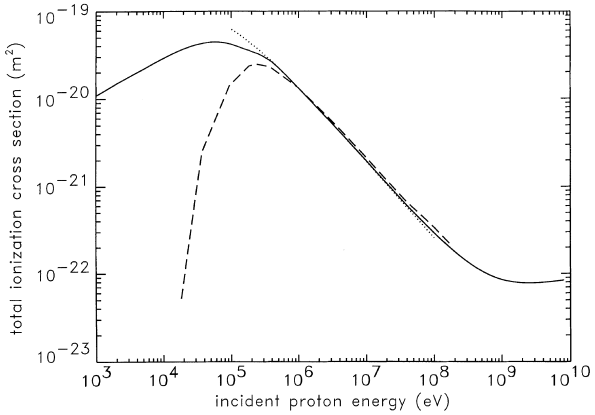

Fig. 6. Total ionisation cross section for protons in liquid water. The solid curve consists of three parts: the semi-empirical formula, as described in Section 3.3 for proton energies below 500 keV, the Born formula for proton energies in the range from 500 keV to 10 MeV, and the relativistic Bethe formula for proton energies exceeding 10 MeV. The dashed curve shows the results of the non-relativistic Born formula which are inadequate at energies below 500 keV and above 10 MeV. Also shown is the electron impact ionisation cross section for electrons (Dingfelder et al., 1998, dashed curve) with the same speed as the protons.

All the cross sections of Fig. 6 are shown again in Fig. 7 in a relativistic Fano plot. The Fano plot is a more sensitive test of the consistency of the cross sections. There, according to Eq. (9), the total ionisation cross section multiplied by the speed squared is plotted versus x (Eq. (10)). For higher speeds the cross section shows the asymptotic straight line, as expected. The deviation from the straight line for lower speeds are clearly seen as well as the failure of the Born-approximation (dotted curve) for low speeds, and also in the relativistic region. The dashed curve, again, shows the recently derived electron cross section at the same speed. All the features of the cross sections discussed in the preceding paragraph are more clearly seen in the Fano plot.

Finally, the differential ionisation cross sections are displayed in Fig. 8 in the form of a Platzman plot, which is a simple, but powerful way graphically to present and evaluate such data (ICRU, 1996). Here, the ratio Y of the energy differential cross section $d\sigma/dE$ to the Rutherford cross section for a single target electron is plotted as a function of Ry/E, where $E = W + I_1$ is the sum of the secondary electron kinetic energy W and the lowest binding energy for liquid water (Dingfelder et al., 1998) $I_1 = 10.79$ eV. Each curve corresponds to a fixed proton energy τ. By choosing Ry/E as a variable on the horizontal axis instead of E, the area under the curve is proportional

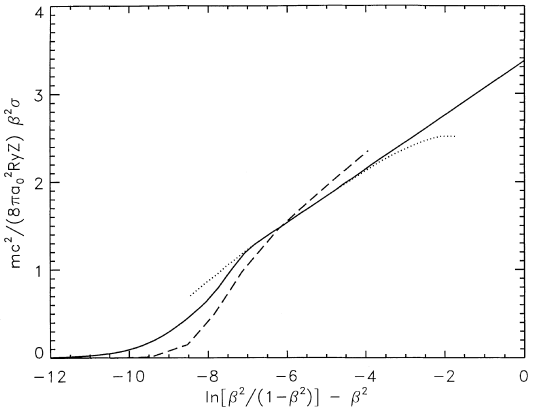

Fig. 7. Relativistic Fano plot of the total ionisation cross section. Shown are the same data as in Fig. 6.

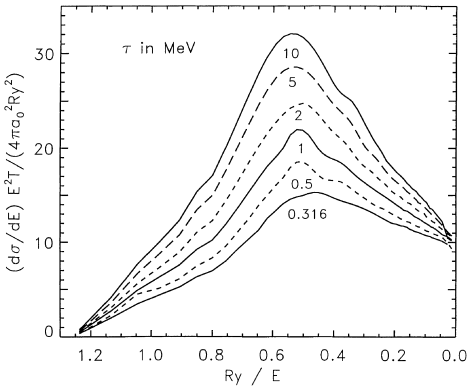

Fig. 8. Platzman plot of the secondary electron spectra for the Born approach. Symbol $E = W + I_1$ represents the secondary electron energy plus the binding energy of the outermost shell, $I_1 = 10.79$ eV. The curves are for different proton energies, from top to bottom for $\tau = 10$, 5, 2, 1 MeV, 500 and 316 keV.

to the total ionisation cross section multiplied by the proton energy τ. The curves show a maximum at around $\mathrm{Ry}/E = 0.55$, which corresponds to a secondary electron energy of $W \approx 14$ eV, and reaches a value of $Z = 10$ for high energy transfers ($\mathrm{Ry}/E \to 0$), which is the number of electrons in the water molecule. The shape is similar to the case of electron impact (Dingfelder et al., 1998, Fig. 4), for which the maximum is at the same energy but narrower and less smooth. The peak at high energy transfers ($\mathrm{Ry}/E \to 0$), seen for electron impact and resulting from the electron exchange contributions, i.e., the indistinguishability of electrons, is missing for proton impact.

3. Slow protons

3.1. Background

As indicated in the last section, the FBA is not valid for describing slow-proton interactions. We call a proton slow, when its kinetic energy is below 500 keV.

Table 1
List of all inelastic processes for slow protons considered in this work[a]

(1) Excitation	$p + H_2O \to p + H_2O^*$
(2) Ionisation	$p + H_2O \to p + H_2O^+ + e^-$
(3a) Charge transfer	$p + H_2O \to H1^* + H_2O^+$
(3b) ... into continuum	$p + H_2O \to (p + e^-) + H_2O^+$
(4) Stripping	$H + H_2O \to p + e^- + H_2O^*$
(5) Ionisation	$H + H_2O \to H + H_2O^+ + e^-$

[a] More explication in the text.

For slower and slower protons other inelastic processes, which are suppressed for higher energies, become more and more important. All the inelastic processes considered in this study are listed in Table 1. Aside from excitation (Table 1, (1)) and ionisation (2), which decrease in importance below 100 keV, charge transfer (3) becomes dominant. In this process an electron from the water molecule is transferred to the moving slow proton, to form a neutral hydrogen atom H, either in any bound state (ground or excited state) (3a) or a continuum state (3b), where the electron is travelling together with the proton at the same speed. After the charge transfer process the neutral hydrogen atom becomes either stripped (4) or ionises (5) a water molecule. In the first case, a proton, and a (possibly excited) water molecule remains, an electron is ejected in forward direction with nearly the same velocity as the proton. In the latter case, the neutral hydrogen, an ionised water molecule and an ejected electron with a similar distribution of electron energy and ejection angle as for protons result. Excitations of the water molecule by the neutral hydrogen atom should occur rarely and are neglected in this study.

At lower incident energies, say below 1 keV, also elastic scattering of the proton by the water molecules becomes important. This process is also called nuclear scattering.

In the following subsections we review all the experiments which we are aware of and which refer largely to water vapour, and present cross section sets suitable to use for liquid water. These sets are obtained taking into account medium phase effects, which have been experimentally seen in stopping cross sections, for which data on the vapour, liquid and ice phase of water exist. The derivation and calculation of the stopping cross sections are described in Section 3.8.

Most of the experiments refer to electron capture and loss processes, while experimental data on ionisation cross sections are scarce. The total cross sections for the different processes are characterised by the initial and the final charge state of the incident particle, and the total cross section for a certain process is written as σ_{ij}, where i indicates the incident charge state and j the final charge state. Therefore, σ_{11} refers to the total ionisation cross section of the proton, σ_{10} to the charge transfer to the proton (also called capture process), and σ_{01} to the stripping of hydrogen (also called electron-loss process).

Note, that for practical use of the following semi-empirical equations of this chapter all variables and parameters of dimension energy are used in units of eV (e.g. primary particle energy τ, binding energies) and all variables and parameters of dimension length in units of meter (e.g. Bohr radius a_0).

3.2. Excitations

To our knowledge, there are no experiments on excitation cross sections for proton impact, neither for water vapour nor for liquid water, apart from the work of Yousif and co-workers (Yousif et al., 1986), who present cross sections for Balmer Hα emission from H_2O through dissociative excitations of the target in the energy range from 5 to 100 keV. Semi-empirical treatments relate proton excitation cross sections to electron excitation cross sections and use systematics of the cross sections for their analytical representation. We know the speed dependence and the very weak mass dependence of the FBA cross sections. The electron excitation cross sections calculated previously for liquid water (Dingfelder et al., 1998) also include low-energy corrections, which account for deviations from the FBA.

After careful deliberations we adopt the following semi-empirical model which uses a speed scaling on the basis of the electron excitation cross sections, together with extensions towards lower proton energies. This idea was developed by Miller and Green (1973), who also presented proton excitation cross sections for water vapour. The cross section for a single excitation level k is represented by the analytic form

$$\sigma_{\mathrm{exc},\,k}^{\mathrm{proton}}(\tau)=\frac{\sigma_0(Za)^{\Omega}(\tau-E_k)^{\nu}}{J^{\Omega+\nu}+\tau^{\Omega+\nu}}, \tag{34}$$

where σ_0 is a constant ($\sigma_0=10^{-20}$ m^2), Z the number of electrons in the target material, and E_k the excitation energy. Symbols ν and Ω are dimensionless parameters and a and J parameters of the dimension of energy (eV).

The parameters a and Ω represent roughly the high-energy limit and are inferred from the speed scaling of electron and proton cross sections in the Born-theory, while J and ν describe the low-energy behaviour and are inferred from the systematics of such parameters from a variety of substances (Green and McNeal, 1972). The parameters J and a are related to the maximum $\sigma_{\max}^{\mathrm{e}}$ of the electron excitation cross section, at an electron energy $T_{\max}$. Thus, J and a are given by

$$J=C_2\frac{M}{m}T_{\max}\left(\frac{\Omega}{\nu}\right)^{1/(\Omega+\nu)} \tag{35}$$

$$a=C_2\frac{M}{m}\frac{T_{\max}}{Z}\left(\frac{C_1\sigma_{\max}^{\mathrm{e}}(\Omega+\nu)}{\sigma_0\nu}\right)^{1/\Omega}, \tag{36}$$

where C_1 and C_2 are parameters connected with Ω and ν. The derivation of Eqs. (35) and (36) is complicated and not unique, as described in detail by Miller and Green (1973). They also made recommendations for the other parameters: $\nu=1$, $C_1=4$, $C_2=0.25$, if $\Omega\approx 1$ and $\nu=2$, $C_1=4$, $C_2=1$, if $\Omega\gg 1$. We choose the parameters a and Ω so that the semi-empirical approach agrees in the high-energy limit with the results obtained from the FBA. In detail, Ω represents the slope of the cross section, and C_1 is used to adjust the magnitude of the cross section for incident proton energies exceeding 500 keV. The parameters ν and C_2 are taken from the recommendations of Miller and Green, discussed above.

The set of parameter values thus determined is displayed in Table 2 for all five excited states considered. The table includes also the values of the cross section maximum $\sigma_{\max}^{\mathrm{e}}$ for electron impact (Dingfelder et al., 1998) and the electron kinetic energy $T_{\max}$ there. Fig. 9 shows the excitation cross section for proton impact, as calculated from Eq. (34) together with the contributions from the different excited states for proton incident energies from a few 10 eV to 10 MeV. Above 500 keV incident proton energy the semi-empirical cross sections agree with the results obtained from the FBA.

Fig. 9 includes the result, shown by the dashed curve, of the straight forward scaling, defined by

$$\sigma_{\mathrm{exc}}^{\mathrm{proton}}(\tau)=\sigma_{\mathrm{exc}}^{\mathrm{electron}}(T),\quad T=\frac{m}{M}\tau. \tag{37}$$

This cross section has been sometimes used in earlier track-structure simulations (see e.g. Zaider et al., 1983), which only considered fast protons. However, the use of Eq. (37) for slow protons leads to the unrealistic drop of the proton excitation cross section,

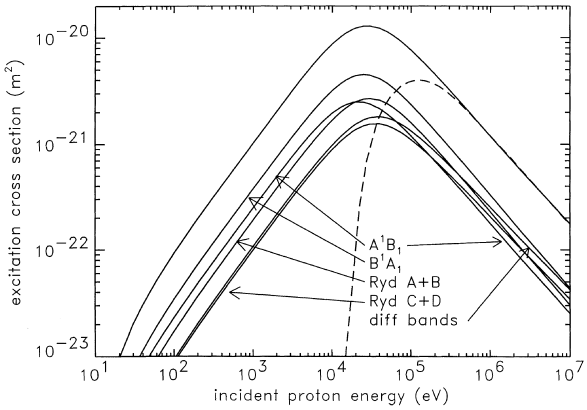

Fig. 9. Total excitation cross section σ_{exc} for liquid water vs. proton incident energy. Shown are the total excitation cross section (top curve) and the contributions of the different excited states, as calculated in the Miller model (proton energies below 500 keV) and the Born formula (proton energies above 500 keV). Also shown is the total electron excitation cross section (dashed curve) for impact of electrons at the same speed.

towards its vanishing at $\tau = 15$ keV, which corresponds to the excitation threshold $E_k = 8.17$ eV for electrons (Dingfelder et al., 1998).

The excitation cross sections for neutral hydrogen impact are neglected in view of lack of experimental or theoretical information and the probable unimportance for our purpose.

3.3. Ionisation by protons

Experiments on proton ionisation cross sections in liquid water do not seem to have been carried out, while measurements in water vapour are scarce. Toburen and Wilson (1977) measured energy and angular distributions of electrons ejected from water vapour by protons in the energy range from 300 keV to 1.5 MeV. They present also single differential cross sections, but not the total ionisation cross section. A calculation of total ionisation cross sections, based on these data, can be found in Kaushik et al. (1981). Additional measurements for higher proton energies are reported by Wilson et al. (1984); a numerical representation is given by Wilson and Nikjoo (1999). Rudd et al. (1985b) presented total ionisation and charge transfer cross sections for water vapour by measuring the gross cross sections for the production of positive and negative charge and calculated the total cross sections (Rudd et al., 1983). Bolorizadeh and Rudd (1986a) presented differential cross sections, differential in scattering angle and secondary-electron ejection energy for slow proton energies (15–150 keV). Gibson and Reid (1987) also measured these cross sections for the single incident energy of 50 keV.

Rudd and co-workers reviewed all experimental and theoretical approaches to total cross sections (Rudd et al., 1985a) and differential cross sections (Rudd et al., 1992). They developed semi-empirical models for the ionisation cross sections and obtained necessary parameters by best fits to all known data. These semi-empirical models reproduce experimental data on differential and total cross sections reasonably well but do not supply realistic information on the partitioning into subshells. Also, they show difficulties in reproducing stopping cross sections. A more recent survey on the secondary-electron spectra from charged particle interactions is published by the ICRU (1996). There, different theoretical and semi-empirical approaches are discussed and parameters and experimental information are given for a wider variety of atoms and molecules.

After thorough considerations we decided to use the following semi-empirical approach based on ideas suggested by Rudd and co-workers (Rudd et al., 1985a, 1992). In detail, the single differential cross section $\mathrm{d}\sigma/\mathrm{d}E$, differential in energy transfer E is given by

$$\frac{\mathrm{d}\sigma}{\mathrm{d}E} = \sum_{\text{all } j} G_j \frac{\mathrm{d}\sigma^j}{\mathrm{d}W_j}, \tag{38}$$

where $W_j = E - I_j$ is the secondary-electron kinetic energy, I_j the ionisation energy of sub-shell j in liquid water and G_j the partitioning factor to adjust the contributions of the different subshells to the results obtained from the Born approximation. In the Rudd model the secondary-electron energy spectrum $\mathrm{d}\sigma_j/\mathrm{d}W_j$ is represented by a combination of low and high energy parts. Thus, the secondary-electron energy distribution for single ionisation of sub-shell j is given by

$$\frac{\mathrm{d}\sigma^j}{\mathrm{d}w} = \frac{S}{B_j} \frac{F_1(v) + wF_2(v)}{(1+w)^3\left[1 + \exp\left\{\alpha(w - w_c)/v\right\}\right]}. \tag{39}$$

where the two functions F_1 and F_2, which consist of different low and high energy parts, are given by

$$\begin{aligned} F_1(v) &= L_1(v) + H_1(v), \\ F_2(v) &= \frac{L_2(v)H_2(v)}{L_2(v) + H_2(v)}, \end{aligned} \tag{40}$$

with

$$L_1(v) = \frac{C_1 v^{D_1}}{1 + E_1 v^{(D_1+4)}}, \qquad H_1 = \frac{A_1 \ln\left(1+v^2\right)}{v^2 + B_1/v^2},$$

$$L_2(v) = C_2 v^{D_2}, \qquad H_2(v) = \frac{A_2}{v^2} + \frac{B_2}{v^4} \tag{41}$$

The dimensionless variables used here are a scaled sec-

Table 2
Parameters for the excitation cross section (Eq. (34)) for liquid water

k	Excited state	E_k (eV)	a (eV)	J (eV)	Ω	ν	C_1	C_2	$T_{\max}$ (eV)	$\sigma^{\mathrm{e}}_{\max}$ (m^2)
1	$\tilde{A}^1B_1$	8.17	876	19820	0.85	1	3.132	0.25	47.15	8.00×10^{-22}
2	$\tilde{B}^1A_1$	10.13	2084	23490	0.88	1	3.468	0.25	54.78	1.30×10^{-22}
3	Ryd $A + B$	11.31	1373	27770	0.88	1	3.324	0.25	65.75	8.11×10^{-22}
4	Ryd $C + D$	12.91	692	30830	0.78	1	3.028	0.25	77.22	5.19×10^{-22}
5	Diffuse bands	14.50	900	33080	0.78	1	3.028	0.25	82.86	6.03×10^{-22}

Table 3
Parameter sets for the differential ionisation cross sections (Eq. (39)) on proton impact for liquid water (this work), water vapour, and K-shell ionisation (Rudd et al., 1992)

Parameter	Liquid	Vapour	K-Shell
A_1	1.02	0.97	1.25
B_1	82.0	82.0	0.50
C_1	0.45	0.40	1.00
D_1	−0.80	−0.30	1.00
E_1	0.38	0.38	3.00
A_2	1.07	1.04	1.10
B_2	14.6	17.3	1.30
C_2	0.60	0.76	1.00
D_2	0.04	0.04	0.00
α	0.64	0.64	0.66

ondary-electron energy $w = W/B_j$, and a scaled speed squared of the particle $v^2 = T/B_j$, where B_j is the binding energy of the sub-shell j. Furthermore, $S = 4\pi a_0^2 N_j(\mathrm{Ry}/B_j)^2$, with N_j the number of electrons in the sub-shell j, $w_c = 4v^2 - 2v - \mathrm{Ry}/(4B_j)$ and α, a numerical parameter related to the relative size of the target molecule.

Note that the symbols in Eqs. (38)–(41) have been taken from Rudd and co-workers (Rudd et al., 1985a, 1992). In particular, C_1, C_2, v and S are not to be confused with the same symbols used earlier in the present paper.

Parameters for many atoms and molecules, including water vapour, can be found in the work by Rudd et al. (1992) and in the ICRU report 55 (ICRU, 1996). The parameter set for liquid water is displayed in Table 3 together with the parameter set for water vapour. Also given there is a special parameter set for the K-shell ionisation (Rudd et al., 1992), common to both phases. The Rudd model gives values of the K-shell ionisation cross section close to the numerical compilation by Paul and Sacher (1989), as seen in Fig. 1, and should be adequate for practical applications.

The parameter set for liquid water has been obtained in the following way: we started with the parameter set for water vapour, and modified it in order to reproduce the recommended ICRU stopping cross section (ICRU, 1993) for liquid water, which is the only relevant experimental information on the liquid phase, taking into account a realistic partitioning into subshells. The partitioning factor G_j is determined by adjusting the shell contributions of the semi-empirical model at higher incident particle energies ($\tau > 500$ keV) to the results of the Born approximation. The obtained values of G_j are displayed in Table 4. The parameter set for water vapour itself does not reproduce the recommended ICRU stopping cross sections for water vapour (ICRU, 1993) very well. Note that the binding energies B_j of the vapour phase are used as parameters in Eq. (39) and cannot be changed or even replaced with those (I_j) of the liquid phase. For completeness the binding energies of both phases are also given in Table 4. In the adoption of the parameter set for liquid we chose to modify the secondary-electron spectra minimally on going from vapour to liquid. The adopted spectra for liquid are slightly less intense (as seen in Fig. 10). This trend is consistent with a general expectation (Inokuti, 1991) that secondary electrons, especially of lower energies, should be suppressed owing to scattering by surrounding molecules in liquid.

Table 4
Ionisation thresholds and number of electrons N for different ionisation sub-shells in water vapour (B_j, Rudd et al., 1992) and liquid water (I_j, Dingfelder et al., 1998). Also given the partitioning factor G_j

Shell j	I_j (eV)	B_j (eV)	N_j	G_j
$1a_1$	539.00	539.70	2	1.00
$2a_1$	32.30	32.20	2	0.52
$1b_2$	16.05	18.55	2	1.11
$3a_1$	13.39	14.73	2	1.11
$1b_1$	10.79	12.61	2	0.99

Fig. 10 shows the ratio of the semi-empirical single differential energy transfer cross section $\mathrm{d}\sigma/\mathrm{d}E$ to the Rutherford cross section $(\mathrm{d}\sigma/\mathrm{d}E)_{\mathrm{Rutherford}} = 4\pi a_0^2 \mathrm{Ry}^2/(TE^2)$ for liquid water (solid curves) and water vapour (dotted curves) for different incident proton energies (from bottom to top $\tau = 15$, 30, 100, 300 keV, and 1 MeV) as a function of the secondary-elec-

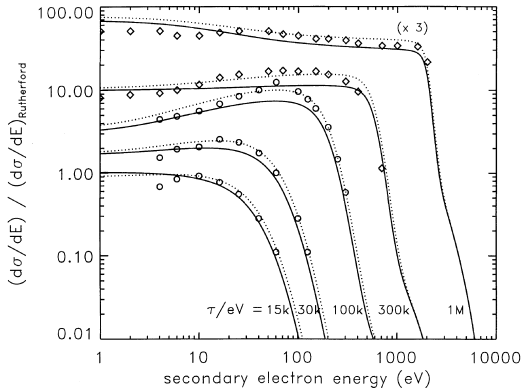

Fig. 10. Ratios of the semi-empirical differential cross sections for liquid water (solid curves) and water vapour (dotted curves) to the Rutherford cross section vs. energy transfer for different incident proton energies (from bottom to top $\tau = 15$, 30, 100, 300 keV, and 1 MeV), compared to experimental data of Bolorizadeh and Rudd (1986a) (circles), and of Toburen and Wilson (1977) (diamonds).

tron energy W. Also shown are experimental data for proton impact on water vapour by Bolorizadeh and Rudd (1986a) (circles) and by Toburen and Wilson (1977) (diamonds).

The total ionisation cross section is obtained by numerically integrating $d\sigma/dE$ over E. Alternatively, a semi-empirical formula for the total ionisation cross section is given by Rudd et al. (1985a), and it leads to values close to those obtained by numerical integration. This formula is also used to represent other total cross sections such as the stripping cross section of hydrogen.

The total cross section is represented by the harmonic mean of a high-energy part σ_{high} and a low-energy part σ_{low}

$$\sigma(\tau) = \left(\frac{1}{\sigma_{\text{low}}} + \frac{1}{\sigma_{\text{high}}}\right)^{-1}. \tag{42}$$

The low-energy part is given as

$$\sigma_{\text{low}}(\tau) = 4\pi a_0^2 \left(C\left(\frac{T}{\text{Ry}}\right)^D + F\right), \tag{43}$$

where C, D and F are parameters, and $T = (m/M)\tau$ represents the kinetic energy of an electron travelling with the same speed as the proton. The high-energy part is given as

$$\sigma_{\text{high}}(\tau) = 4\pi a_0^2 \left(\frac{\text{Ry}}{T}\right)\left[A \ln\left(1 + \frac{\text{Ry}}{T}\right) + B\right], \tag{44}$$

where A and B are parameters. (Note that the symbols in Eqs. (42)–(44) follow Rudd et al. (1985a). In particular, A, B, C, D and F are all parameters, and are not to be confused with our $A(E)$, $B(E)$, etc.)

Parameters for total ionisation cross sections can be found for various atoms and molecules in the review article of Rudd et al. (1985a), and for water vapour in Rudd et al. (1985b). Parameter sets used in the present work are displayed in Table 5.

Fig. 11 presents total ionisation cross sections obtained by numerical integration of the differential cross sections. Shown are results of the present calculation for proton impact (solid curve) and hydrogen impact (dashed curve), to be further discussed in Section 3.6, on liquid water, the recommended cross section for proton impact on water vapour (Rudd et al., 1992) (dash–dotted curve) and various experimental data from Rudd et al. (1985b) (symbols). The liquid water cross section is somewhat smaller than the vapour cross section at high incident energies and around the maximum, which is shifted to a little lower incident energies, but is comparable or even larger at lower incident energies. Indeed, the maximum of the liquid water stopping cross section (to be discussed further in Section 3.7) is smaller in its magnitude and width, which indicates that the cross section receives less contributions around the maximum, and more contributions at lower energies again.

Table 5
Parameter sets for total cross sections (Eq. (42)). σ_+ and σ_- are taken from Rudd et al. (1985b)

	σ_-	σ_+	σ_{01}
A	2.98	2.98	2.835
B	4.42	4.42	0.310
C	1.48	1.48	2.100
D	0.75	0.75	0.760
F	–	4.80	–

3.4. Charge transfer

In contrast to the very few measurements of ionisation cross sections there is a good data base for charge-changing events. A recent review article on newer data for various molecules is published by Toburen (1998); an older one by Tawara (1978) is a good data source on cross sections of various atoms and molecules, too. There are many measurements of the electron-capture cross section σ_{10} in water vapour, but there is no experimental corresponding information on the liquid or solid phase. Unfortunately, the absolute values of the cross sections measured by different workers differ by up to one order of magnitude. Therefore, in a most recent measurement, Lindsay et al. (1997) tried to confirm or reject earlier measurements.

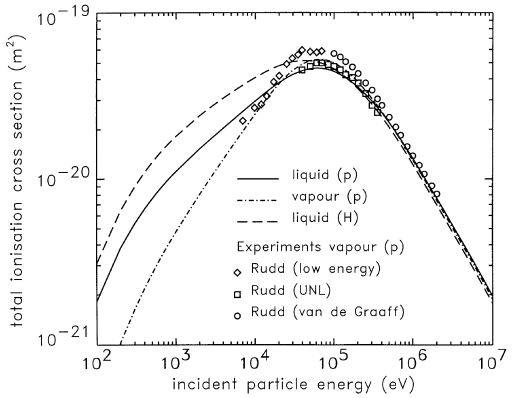

Fig. 11. Total ionisation cross sections vs. incident particle energies. Shown are the proton ionisation cross sections (solid curve) and the hydrogen ionisation cross section (dashed curve) for liquid water and the proton ionisation cross section (dashed–dotted curve) for water vapour, compared to various measurements (different symbols) for proton impact on water vapour (Rudd et al., 1985b).

They measured the angular differential and total charge-transfer cross sections at proton energies of 500 eV, 1.5 and 5 keV and found good agreement with the measurements by Dagnac et al. (1970) (concerning charge-transfer and electron-loss cross sections for proton energies in the range from 2 to 60 keV) and by Berkner et al. (1970) (concerning charge-transfer cross sections for proton energies in the range from 150 eV to 5 keV), while the data of Chambers (cited by Lindsay et al., 1997) (in the energy range from 1 to 10 keV) are somewhat higher than the new measurements. They discarded the data by Coplan and Ogilvie (1970) for the proton energy range from 100 to 500 eV, which are higher by a factor of 3, and the data of Koopman (1968) for the proton energy range from 40 eV to 1.5 keV, which are about an order of lower magnitude.

Other data are due to measurements by Cable (cited by Koopman, 1968) for protons between 150 and 500 eV and by Toburen et al. (1968) (concerning charge-transfer and electron-loss cross sections for an energy range from 100 keV to 2.5 MeV), which is the only data source for higher proton energies.

Rudd and co-workers (Rudd et al., 1985b) measured the production cross section of positive (σ_+) and negative (σ_-) charges, performing separate measurements with different accelerators in the total energy range from 7 keV up to 4 MeV, and fitted them to their semi-empirical model Eqs. (42)–(44) for total cross sections. The charge-transfer cross section is then $\sigma_{10} = \sigma_+ - \sigma_-$; the parameter set is displayed in Table 5. The resulting cross section agrees well with the experimental data for proton energies of 1–100 keV, but is clearly higher than the experimental data of Toburen et al. (1968) for energies above 100 keV on one hand, and leads to a constant value for energies below 1 keV on the other hand, while most of the other experimental values increase with decreasing proton energies. This probably means that the simple semi-empirical model is not flexible enough to account for details. In particular, the determination of σ_{10} by the subtraction is not very precise when σ_+ and σ_- are comparable.

Consequently we decided to represent the charge-transfer cross section by an analytic formula, consisting of straight lines for low and high proton energies on a doubly logarithmic scale, both connected by a power law. In detail, the charge transfer cross section is expressed by

$$\sigma_{10}(\tau) = 10^{Y(X)}, \tag{45}$$

where

$$X = \log(\tau), \quad \tau \text{ in eV}, \tag{46}$$

and

Table 6
Parameter set for the charge transfer cross section (Eq. (45))

a_0	−0.180
b_0	−18.22
c_0	0.215
d_0	3.550
a_1	−3.600
b_1	−1.997
x_0	3.450
x_1	5.251

$$Y(X) = \left[a_0 X + b_0 - c_0 (X - x_0)^{d_0} \Theta(X - x_0)\right]\Theta(x_1 - X) + (a_1 X + b_1)\Theta(X - x_1), \tag{47}$$

where $\Theta(x)$ represents the Heaviside step function. The parameters a_0 and b_0 determine the low-energy straight line, a_1 and b_1 the high-energy one, c_0 and d_0 the power law in between, connected to the low-energy straight line at x_0. The connection point x_1 to the high-energy line and the parameter b_1 are calculated as

$$x_1 = \left(\frac{a_0 - a_1}{c_0 d_0}\right)^{1/(d_0 - 1)} + x_0, \tag{48}$$

$$b_1 = (a_0 - a_1)x_1 + b_0 - c_0 (x_1 - x_0)^{d_0}, \tag{49}$$

using the first derivative.

The parameters were chosen by considering the experimental data on water vapour as a basis, i.e., so that σ_{10} is close to the data of Lindsay et al. (1997) and Dagnac et al. (1970) for low and medium energies and of Toburen et al. (1968) for higher energies, and by adjusting the contributions of the cross section to the total stopping cross section in respect to the recommended values for the liquid phase. The charge transfer cross section mainly influences the probability of finding the projectile in a definite charge state, but also the contribution of the charge changing process. The parameter set is displayed in Table 6; the charge transfer cross section (solid curve) is plotted in Fig. 12 and compared to various experimental data. The figure also shows the semi-empirical model of Rudd et al. (1985b) for water vapour (dash–dotted curve) as described above.

3.5. Stripping of neutral hydrogen

Dagnac et al. (1970) measured the stripping cross section for lower energies, while Toburen et al. (1968) presented data for higher energies. Both groups also review theoretical treatments mostly concerning the hydrogen target. Because there is no satisfactory theoretical treatment of water we decided to use again the

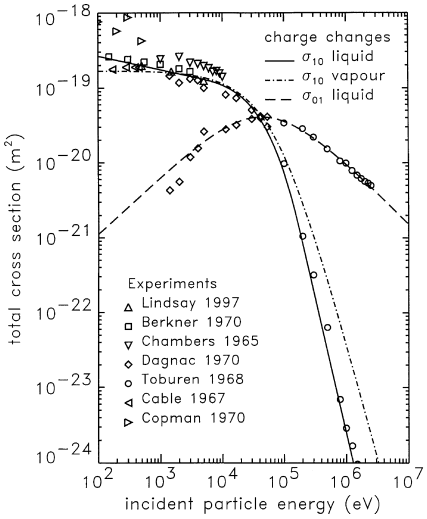

Fig. 12. Charge-changing cross sections for liquid water vs. incident particle energy. Shown are the charge transfer cross section σ_{10} (solid curve) and the electron loss cross section σ_{01} (dashed curve), compared to various experiments for water vapour (symbols). Also shown is the charge transfer cross section for water vapour (Rudd et al., 1985b), dash–dotted curve.

semi-empirical formulas, Eqs. (42)–(44), and fit the parameters to these existing experiments. We again use the freedom of adjusting the parameters to reproduce the stopping cross section. The charge-transfer cross section and the stripping (or electron-loss) cross section together determine the probability of finding the projectile either in the "proton" state or in the "hydrogen" state. Both cross sections together determine the contribution of the charge-changing process and influence the total stopping cross section. The parameters obtained are shown in Table 5; the stripping cross section itself is shown as the dashed curve in Fig. 12.

3.6. Ionisation by neutral hydrogen

For fast collisions, hydrogen-impact cross sections differ from proton-impact cross sections by the effects of screening of the nuclear charge by the bound hydrogen electron, by the contributions to the electron spectra from the stripping of the hydrogen electron and by contributions from electron–electron interactions of the hydrogen electron with the target electrons. The relative importance of these three processes depends on the incident particle energy, the ejected electron energy and the ejected electron angle. Unfortunately, there seems to be no experimental information on these processes for the liquid or solid phase. Fortunately, there are data for proton- and hydrogen-impact in gas targets (IAEA, 1995; Toburen, 1998), including water vapour (Bolorizadeh and Rudd, 1986a, 1986b). Bolorizadeh and Rudd measured angular and energy differential cross sections for proton (Bolorizadeh and Rudd, 1986a) and hydrogen (Bolorizadeh and Rudd, 1986b) impact on water vapour and presented the ratio of the single differential cross section for hydrogen impact to that for proton-impact as a function of the secondary-electron energy (cf. Fig. 5 of Bolorizadeh and Rudd, 1986b). This ratio is almost independent of the secondary-electron energy for a fixed incident particle energy, except for the contributions from the stripping of the hydrogen electron, i.e., the broad peak at the electron velocity nearly equal to the projectile velocity. The ratio is about 2.0 for 20-keV incident energy and decreases down to 0.9 for 100-keV incident energy.

Therefore, after thorough considerations we decided to use the same secondary-electron spectrum for proton impact as a starting point, but modified by a function $g(\tau)$ that depends only on the incident particle energy τ and not on the energy transfer E. On the basis of data (Bolorizadeh and Rudd, 1986b; Toburen, 1998) we adopt the function $g(\tau)$ in the following way. At low incident particle energies, $g(\tau)$ is a constant higher than unity. At higher incident energies, $g(\tau)$ is slightly lower than unity, in view of the screening of the nuclear charge by the bound electron in hydrogen. Thus, the energy-transfer differential cross section for ionisation by hydrogen impact is given by

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}E}\right)_{\text{hydrogen}} = g(\tau)\left(\frac{\mathrm{d}\sigma}{\mathrm{d}E}\right)_{\text{proton}}, \tag{50}$$

with

$$g(\tau) = 0.8\left(1+\exp\left[\frac{\log(\tau)-4.2}{0.5}\right]\right)^{-1}+0.9. \tag{51}$$

The parameters are determined by considering the experimental data on water vapour as a basis and by adjusting the contribution of the cross section to the recommended total stopping cross section in liquid water by ICRU (1993). Consequently, the total ionisation cross section for hydrogen impact, shown by the dashed curve in Fig. 11, is larger by a factor of 1.7 than the proton ionisation cross section for low incident particle energies and by a factor of 0.9 smaller for high incident particle energies.

3.7. Phase effects

Some information of the influence of the state of aggregation (phase effect) to the cross sections can be obtained from electronic stopping cross sections. The stopping cross sections σ_{st} are of interest in many fields of science dealing with radiation. Therefore, the ICRU

published recommended stopping cross sections for most of the molecules and materials of interest to radiation science (ICRU, 1993), which are based on both theoretical and experimental information. Because of the relevance of water in medical applications of ionising radiation (i.e. radiotherapy) already in 1952 measurements with protons as incident particles were performed (Wenzel and Whaling, 1952; Reynolds et al., 1953). Also some theoretical work has been done to calculate stopping cross sections for water explicitly in different phases (Xu et al., 1985; Fainstein et al., 1996).

Recent years have seen extensive studies on phase effects in several materials, for instance, on the energy loss of protons in gaseous and solid zinc targets (Bauer et al., 1992; Arnau et al., 1994) as well as in water vapour and ice (Mitterschiffthaler and Bauer, 1990; Bauer et al., 1994; Bauer et al., 1997). A notable phase effect was found in these studies between water vapour and ice in the energy range around the stopping cross section maximum. The maximum of the stopping cross section of ice is smaller in the absolute value and narrower than the maximum of water vapour, and it is shifted to somewhat higher proton energies compared to the vapour phase. At lower particle energies, the phase effect amounts to as much as to 15% at 20 keV. We see that $\sigma_{\mathrm{st}}^{(\mathrm{vapour})} > \sigma_{\mathrm{st}}^{(\mathrm{ice})} > \sigma_{\mathrm{st}}^{(\mathrm{liquid})}$ in the maximum using the ICRU data for $\sigma_{\mathrm{st}}^{(\mathrm{liquid})}$ (ICRU, 1993).

At higher proton energies the phase effect is small. Here, the influence of charge-changing processes is negligible. Therefore, we can use the Bethe theory to estimate the phase effect (Bauer et al., 1994) as

$$\sigma_{\mathrm{st}}^{(\mathrm{phase\ 1})} - \sigma_{\mathrm{st}}^{(\mathrm{phase\ 2})} = \frac{4\pi Z\, e^4}{mv^2} \ln\left(\frac{I_{\mathrm{phase\ 2}}}{I_{\mathrm{phase\ 1}}}\right), \tag{52}$$

where I is the mean excitation energy of each phase. The recommended values (ICRU, 1984) are $I_{\mathrm{vapour}} = 71.6$ eV, $I_{\mathrm{ice}} = 75.0$ eV and $I_{\mathrm{liquid}} = 75.0$ eV, newer experimental values are $I_{\mathrm{ice}} = 80.9$ eV (Bauer et al., 1994) and $I_{\mathrm{liquid}} = 79.7$ eV (Bichsel and Hiraoka, 1992). Our model (Dingfelder et al., 1998) gives $I_{\mathrm{liquid}} = 81.8$ eV. Using the recommended values the difference at 300 keV is $\sigma_{\mathrm{st}}^{(\mathrm{vapour})} - \sigma_{\mathrm{st}}^{(\mathrm{liquid/ice})} = 3.7 \times 10^{-20}$ eV m^2, corresponding to about 2% (Bauer et al., 1994).

3.8. Stopping cross sections

Ions moving in a medium can capture and lose electrons and thus are in a dynamic charge equilibrium. The electron loss and capture processes will contribute to the stopping cross section considerably and the interaction of the ion in different charge states will influence the total stopping cross section.

Theoretically, the stopping cross section σ_{st} for a charged ion in a medium is described by the charge state approach (Bates and Griffing, 1953; Allison and Warshaw, 1953), where σ_{st} is a sum over all charged states i of the projectile

$$\sigma_{\mathrm{st}} = \sum_{i,\, j,\, i \neq j} \Phi_i \left(\sigma_{\mathrm{st},i} + \sigma_{ij} T_{ij}\right). \tag{53}$$

Here, Φ_i denotes the probability to find the ion in the charge state i, $\sigma_{\mathrm{st},i}$ the (electronic) stopping cross section of charge state i, σ_{ij} the charge-changing cross section from charge state i to charge state j (capture and loss of electrons, respectively) and T_{ij} the energy loss connected with the charge changing event. We consider only one electron capture and loss processes and neglect multi-electron processes, in the present work. Therefore, our system contains two different charge states of the projectile: $i = 0$: neutral hydrogen H and $i = 1$: proton p. Therefore, Eq. (53) reduces to

$$\sigma_{\mathrm{st}} = \Phi_0 \left(\sigma_{\mathrm{st},\,0} + \sigma_{01} T_{01}\right) + \Phi_1 \left(\sigma_{\mathrm{st},\,1} + \sigma_{10} T_{10}\right). \tag{54}$$

A lower bound of the sum $T_{01} + T_{10}$, i.e., the energy loss per charge-changing cycle, is given by the sum of the ionisation energy of the water molecule I_0 and the kinetic energy of the electron, when travelling with the speed v of the proton, $T = mv^2/2$,

$$T_{01} + T_{10} = I_0 + \frac{mv^2}{2}. \tag{55}$$

The probabilities Φ_0 and Φ_1 are

$$\Phi_0 = \frac{\sigma_{10}}{\sigma_{01} + \sigma_{10}}, \quad \text{and} \quad \Phi_1 = \frac{\sigma_{01}}{\sigma_{01} + \sigma_{10}}. \tag{56}$$

Therefore, we can write

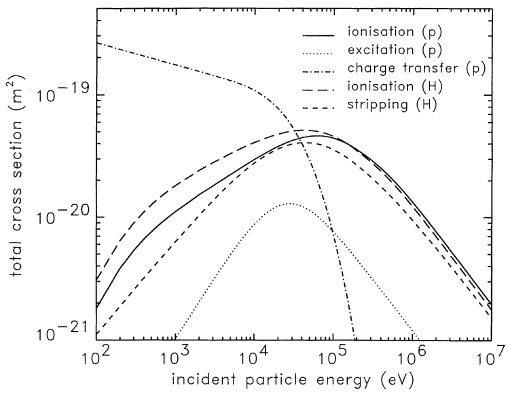

Fig. 13. Total cross sections for the different processes in liquid water: proton ionisation σ_{11} (solid curve), hydrogen ionisation σ_{00} (long dashed curve), charge transfer σ_{10} (dash–dotted curve), electron loss σ_{01} (short dashed curve), and proton excitation cross section (dotted curve).

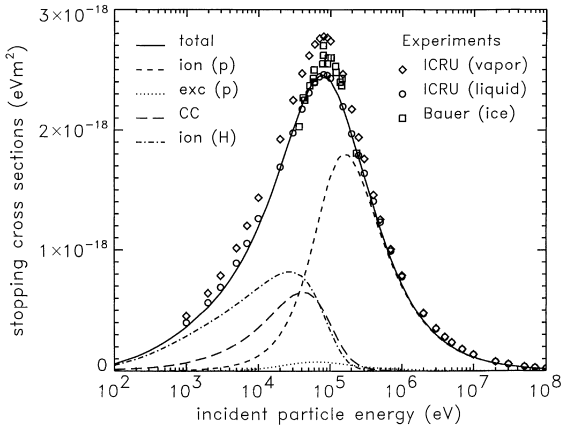

Fig. 14. Stopping cross section σ_{st} for liquid water (solid curve) with contributions from proton ionisation (short dashed curve), hydrogen ionisation (dash–dotted curve), charge changing processes (long dashed curve) and proton excitation (dotted curve), compared with experimental and recommended data for the different water phases: liquid water (ICRU, 1993, circles), water vapour (ICRU, 1993, diamonds), and ice (Bauer et al., 1994, squares).

$$\sigma_{st} = \Phi_0\sigma_{st,0} + \Phi_1\sigma_{st,1} + \sigma_{st,CC} \tag{57}$$

with the stopping cross sections $\sigma_{st,\,i}$ of charge state i

$$\sigma_{st,i} = \int_{\text{all } E} E\frac{d\sigma_{ii}}{dE}\,dE \tag{58}$$

and the charge-changing stopping cross section $\sigma_{st,CC}$

$$\sigma_{st,CC} = \frac{\sigma_{01}\sigma_{10}}{\sigma_{01}+\sigma_{10}}(I_0 + T). \tag{59}$$

3.9. Results

The total stopping cross sections of the vapour and the liquid phase are used to adjust parameter sets for the different processes and to adopt them for the liquid phase.

Fig. 13 shows all total cross sections together: the proton excitation cross section σ_1^{exc} (dotted curve), the proton ionisation cross section σ_{11} (solid curve), the hydrogen ionisation cross section σ_{00} (long dashed curve), the charge-transfer cross section σ_{10} (dash–dotted curve) and the electron-loss cross section σ_{01} (short dashed curve). One recognises that the charge-transfer cross section dominates for proton energies below, say 30 keV, increasing with decreasing energies, while the ionisation cross sections and the energy-loss cross sections, all similar in shape and order of magnitude, decrease in this energy range. The hydrogen-impact ionisation cross section is larger than the proton-impact ionisation cross section for energies below 100 keV. The electron-loss cross section is always smaller than the ionisation cross sections. The excitation cross section follows more or less the ionisation cross section in shape and is one order of magnitude smaller.

The stopping cross section calculated from these cross sections is shown in Fig. 14 (solid curve) together with the contributions from proton-impact ionisation $\Phi_1\sigma_{st,1}$ (short dashed curve), hydrogen-impact ionisation $\Phi_0\sigma_{st,0}$ (long dashed curve), charge-changing processes $\sigma_{st,CC}$ (dash–dotted curve), and from proton-impact excitations (dotted curve) compared to the recommended ICRU stopping cross section for liquid water (ICRU, 1993) (circles). Also shown are the recommended ICRU stopping cross sections for water vapour (ICRU, 1993) (diamonds) and the experimental stopping cross sections for ice (Bauer et al., 1994) (squares). Since we used the stopping cross sections to normalize and to adjust parameter sets for different processes, our model calculation shows an excellent agreement with the recommended values. At high energies only the proton ionisation cross section contributes appreciably to the stopping cross section; below 300 keV contributions from all other processes appear. The stopping cross section near the maximum depends on all contributions. Below 20 keV it is dominated by the hydrogen ionisation contribution. The contribution from excitations is less than a few percent.

4. Concluding remarks

The present paper reports the results of our work on a self-consistent set of proton inelastic scattering cross sections for liquid water for a wide energy range from very slow protons (100 eV) to highly relativistic ones (10 GeV). This set includes excitation and ionisation cross sections of protons, but also charge transfer, and ionisation and electron loss of moving neutral hydrogen atoms. The data can be used, for example, in track-structure analysis and other quantitative modelling studies in radiation physics, chemistry and biology[2].

The present article is the third report on our continuing efforts toward establishing the best possible set of cross-section data for liquid water, following up Dingfelder et al. (1998) and Dingfelder and Inokuti (1999a), which primarily concern electron interactions. We now plan to report soon on those aspects of proton interactions, such as elastic scattering by a water molecule as a whole and the angular distribution of

[2] A copy of our computer code (in C) generating the present cross sections can be transmitted in an electronically readable form upon request to dingfelder@gsf.de

secondary electrons, which are crucial to the determination of the spatial structure of tracks. Eventually we will extend our work to treat heavier ions of importance to radiation biology and medicine, e. g., α-particles as well as carbon, nitrogen and oxygen ions.

Acknowledgements

We thank L.H. Toburen, P. Bauer and P. Jacob for helpful discussions including valuable ideas to different parts of this paper. The present work is supported by the European Community under Contract No. FI4P-CT95-0011 "Biophysical Models for the Induction of Cancer by Radiation" and in part by the Department of Energy, Office of Science, Nuclear Physics Division, under Contract No. W-31-109-Eng-38.

References

Allison, S.K., Warshaw, S.D., 1953. Passage of heavy particles through matter. Rev. Mod. Phys. 25, 779–817.

Arnau, A., Bauer, P., Kastner, F., Salin, A., Ponce, V.H., Fainstein, P.D., Echenique, P.M., 1994. Phase effect in the energy loss of hydrogen projectiles in zinc targets. Phys. Rev. B 49, 6470–6480.

Ashley, J.C., 1982a. Stopping power of liquid water for low-energy electrons. Radiat. Res. 89, 25–31.

Ashley, J.C., 1982b. Density effect in liquid water. Radiat. Res. 89, 32–37.

Bak, J.F., Meyer, F.E., Petersen, J.B.B., Uggerhøj, E., Østergaard, K., Møller, S.P., Sørensen, A.H., Siffert, P., 1983. New experimental investigations of the density effect on inner shell excitation and energy loss. Phys. Rev. Lett. 51, 1163–1166.

Bak, J.F., Petersen, J.B.B., Uggerhøj, E., Østergaard, K., Møller, S.P., Sørensen, A.H., 1986. Influence of transition radiation and density effect on atomic K-shell excitation. Physica Scripta 33, 147–155.

Bates, D.R., Griffing, G.W., 1953. Inelastic collisions between heavy particles I: Excitation and ionisation of hydrogen atoms in fast encounters with protons and with other hydrogen atoms. Proc. Phys. Soc. London, Sect. A 66, 961–971.

Bauer, P., Kastner, F., Arnau, A., Salin, A., Fainstein, P.D., Ponce, V.H., Echenique, P.M., 1992. Phase effect in the energy loss of H projectiles in Zn targets: experimental evidence and theoretical explanation. Phys. Rev. Lett. 69, 1137–1139.

Bauer, P., Käferböck, W., Nečas, V., 1994. Investigation of the electronic loss of hydrogen ions in H_2O: influence of the state of aggregation. Nucl. Instr. Meth. 93, 132–136.

Bauer, P., Gosler, R., Aumayr, F., Semrad, D., Arnau, A., Zarate, E., Diez–Muiño, R., 1997. Contribution of valence electrons to the electronic energy loss of hydrogen ions in oxides. Nucl. Instr. Meth. B 125, 102–105.

Berkner, K.H., Pyle, R.V., Stearns, J.W., 1970. Cross sections for electron capture by 0.3 to 70 keV deuterons in H_2, H_2O, CO, CH_4, and C_8F_{16} gases. Nucl. Fus. 10 (1970), 145–149.

Bethe, H., 1930. Zur Theoorie des Durchgangs schneller Korpuskularstrahlen durch Materie. Ann. Phys. (Leipzig) 5, 325–400.

Bethe, H., Ashkin, J., 1956. Segrè, E. (Ed.), Experimental Nuclear Physics, vol I. Wiley, New York, p. 166.

Bethe, H.A., Jackiw, R.W., 1997. Intermediate Quantum Mechanics Addison. Wesley, Reading, MA Reprint Series.

Bichsel, H., Hiraoka, T., 1992. Energy loss of 70 MeV protons in elements. Nucl. Instr. Meth. 66, 345–351.

Bichsel, H., Inokuti, M., 1998. Difference in stopping power for protons and deuterons of a given speed. Nucl. Instr. Meth. B 134, 161–164.

Bolorizadeh, M.A., Rudd, M.E., 1986a. Angular and energy dependence of cross sections for ejection of electrons from water vapor. II. 15–150 keV proton impact. Phys. Rev. A 33, 888–892.

Bolorizadeh, M.A., Rudd, M.E., 1986b. Angular and energy dependence of cross sections for ejection of electrons for water vapor. III. 20–150 keV neutral-hydrogen impact. Phys. Rev. A 33, 893–896.

Coplan, M.A., Ogilvie, K.W., 1970. Charge exchange for H^+ and H_2^+ in H_2O, CO_2, and NH_3. J. Chem. Phys. 52, 4154–4160.

Dagnac, R., Blanc, D., Molina, D., 1970. A study on the collision of hydrogen ions H_1^+, H_2^+ and H_3^+ with a water–vapour target. J. Phys. B 3, 1239–1251.

Dingfelder, M., Hantke, D., Inokuti, M., 1997. Interaction cross sections for electron inelastic scattering in liquid water. In: Goodhead, D.T., Neill, P.O, Menzel, H. (Eds.), Microdosimetry: An Interdisciplinary Approach. The Royal Society of Chemistry, Cambridge, pp. 23–26.

Dingfelder, M., Hantke, D., Inokuti, M., Paretzke, H.G., 1998. Electron inelastic-scattering cross sections in liquid water. Radiat. Phys. Chem. 53, 1–18.

Dingfelder, M., Inokuti, M., 1999a. The Bethe surface of liquid water. Radiat. Environ. Biophys. 38, 93–96.

Dingfelder, M., Inokuti, M., 1999b. Fermi density effect on the cross section for a fixed energy transfer, with application to liquid water. In: International Conference on Atomic Collisions in Solids ICACS–18, Book of Abstracts, Odense, Denmark, August 3–8, 48.

Fainstein, P.D., Olivera, G.H., Rivarola, R.D., 1996. Theoretical calculations of the stopping power for protons traversing H, He and simple molecular targets. Nucl. Instr. Meth. B 107, 19–26.

Fano, U., 1956. Differential inelastic scattering of relativistic charged particles. Phys. Rev. 102, 385–387.

Fano, U., 1963. Penetration of protons, alpha particles, and mesons. Ann. Rev. Nucl. Sci. 13, 1–66.

Fermi, E., 1940. The ionisation loss of energy in gases and in condensed materials. Phys. Rev. 57, 485–493.

Friedland, W., Jacob, P., Paretzke, H.G., Merzagora, M., Ottolenghi, A., 1999. Simulation of DNA fragment distributions after irradiation with photons. Radiat. Environ. Biophys. 38, 39–47.

Gibson, D.K., Reid, I.D., 1987. Energy and angular distributions of electrons ejected from various gases by 50 keV protons. Radiat. Res. 112, 418–425.

Grande, P.L., Schiwietz, G., 1992. Energy loss of slow ions:

one-band calculation for alkaline metals. Phys. Lett. A 163, 439–446.

Green, A.E.S., McNeal, R.J., 1972. Analytic cross sections for inelastic collisions of protons and hydrogen atoms with atomic and molecular gases. J. Geophys. Res. 76, 133–144.

Hayashi, H., Watanabe, N., Udagawa, Y., Kao, C.-C., 1998. Optical spectra of liquid water in vacuum uv region by means of inelastic X-ray scattering spectroscopy. J. Chem. Phys. 108, 823–825.

Heller, J.M., Hamm Jr, R.N., Birkhoff, R.D., Painter, L.R., 1974. Collective oscillation in liquid water. J. Chem. Phys. 60, 3483–3486.

Inokuti, M., 1971. Inelastic collisions of fast charged particles with atoms and molecules — the Bethe theory revisited. Rev. Mod. Phys. 43, 297–347 (See also addenda: Inokuti, M., Itikawa, Y., Turner, J.E. 1978. Rev. Mod. Phys. 50, 23–35.

Inokuti, M., Smith, D.Y., 1982. Fermi density effect on the stopping power of metallic aluminium. Phys. Rev. B 25, 61–66.

Inokuti, M., 1991. How is radiation energy absorption different between the condensed phase and the gas phase? Radiat. Effects Defects Solids 117, 143–162.

Inokuti, M. (Ed.), 1994. Advances in Atomic, Molecular and Optical Physics, Vol. 33: Cross-Section Data. Academic Press, Boston.

International Atomic Energy Agency, 1995. Atomic and Molecular Data for Radiation Therapy and Related Research. IAEA–TECDOC–799, IAEA, Vienna.

International Commission on Radiation Units and Measurements, 1984. Stopping Powers for Electrons and Positrons. ICRU Report 37. International Commission on Radiation Units and Measurements, Bethesda, Maryland.

International Commission on Radiation Units and Measurements, 1989. Tissue Substitutes in Radiation Dosimetry and Measurements. ICRU Report 44. International Commission on Radiation Units and Measurements, Bethesda, Maryland.

International Commission on Radiation Units and Measurements, 1993. Stopping Powers and Ranges for Protons and Alpha Particles. ICRU Report 49. International Commission on Radiation Units and Measurements, Bethesda, Maryland.

International Commission on Radiation Units and Measurements, 1996. Secondary Electron Spectra from Charged Particle Interactions. ICRU Report 55. International Commission on Radiation Units and Measurements, Bethesda, Maryland.

Jackson, J.D., 1998. Classical Electrodynamics, 3rd ed. Wiley, New York (Chapter 13).

Kaushik, Y.D., Khare, S.P., Kumar, A., 1981. Inelastic collisions of protons with water vapour. Physica 106 C, 128–134.

Kimura, M., Lane, N.F., 1990. The low-energy, heavy-particle collisions — a close-coupling treatment. In: Bates, D., Bederson, B. (Eds.), Advances in Atomic, Molecular and Optical Physics, vol. 26. Academic Press, Boston, pp. 80–160.

Koopman, D.W., 1968. Light-ion charge exchange in atmospheric gases. Phys. Rev. 166, 57–62.

Landau, L.D., Lifshitz, E.M., 1970. In: Sykes, J.B., Bell, J.S., Kearsley, M.J. (Eds.), Electrodynamics of Continuous Media, 2nd ed. Pergamon Press, London.

Lindsay, B.G., Sieglaff, D.R., Smith, K.A., Stebbings, R.F., 1997. Charge transfer of 0.5, 1.5, and 5 keV protons with H_2O: Absolute differential and integral cross sections. Phys. Rev. A 55, 3945–3946.

Long, K.A., Paretzke, H.G., 1991. Comparison of calculated cross sections for secondary electron emission from a water molecule and clusters of water molecules by protons. J. Chem. Phys. 95, 1049–1056.

Meyerhof, W.E., Jensen, D.G., Kawall, D.M., Kuhn, S.E., Spooner, D.W., Meziani, Z.E., Faust, D.N., 1992. Density effect in Cu K-shell ionisation by 5.1-GeV electrons. Phys. Rev. Lett. 68, 2293–2296.

Miller, J.H., Green, A.E.S., 1973. Proton energy degradation in water vapor. Radiat. Res. 54, 343–363.

Mitterschiffthaler, C., Bauer, P., 1990. Stopping cross section of water vapor for hydrogen ions. Nucl. Instr. Meth. B 48, 58–60.

Paretzke, H.G., 1987. Radiation track structure theory. In: Freeman, G.R. (Ed.), Kinetics of Non-Homogeneous Processes. Wiley, New York, pp. 89–170.

Paul, H., Sacher, J., 1989. Fitted empirical reference cross sections for K–shell ionisation by protons. At. Data Nucl. Data Tables 42, 105–156.

Press, W.H., Teukolsky, S.A., Vetterling, W.T., Flannery, B.P., 1992. Numerical Recipes in Fortran, 2nd ed. Cambridge University Press, Cambridge.

Reynolds, H.K., Dunbar, D.N.F., Wenzel, W.A., Whaling, W., 1953. The stopping cross section of gases for protons, 30–600 keV. Phys. Rev. 92, 742–748.

Ritchie, R.H., Hamm, R.N., Turner, J.E., Wright, H.A., Bolch, W.E., 1991. Radiation interactions and energy transport in the condensed phase. In: Glass, W.A., Varma, M.N. (Eds.), Physical and Chemical Mechanisms in Molecular Radiation Biology. Plenum Press, New York, pp. 99–133.

Rudd, M.E., DuBois, R.D., Toburen, L.H., Ratcliffe, C.A., Goffe, T.V., 1983. Cross sections for ionisation of gases by 5–4000 keV protons and for electron capture by 5–150 keV protons. Phys. Rev. A 28, 3244.

Rudd, M.E., Kim, Y.-K., Madison, D.H., Gallagher, J.W., 1985a. Electron production in proton collisions with atoms and molecules: total cross sections. Rev. Mod. Phys. 57, 965.

Rudd, M.E., Goffe, T.V., DuBois, R.D., Toburen, L.H., 1985b. Cross sections for ionisation of water vapor by 7–4000 keV protons. Phys. Rev. A 31, 492–494.

Rudd, M.E., Kim, Y.-K., Madison, D.H., Gay, T.J., 1992. Electron production in proton collisions with atoms and molecules: energy distributions. Rev. Mod. Phys. 64, 441–490.

Schiwietz, G., 1990. Coupled-channel calculation of stopping powers for intermediate-energy light ions penetrating atomic H and He targets. Phys. Rev. A 42, 296–306.

Slater, J.C., 1930. Atomic shielding constant. Phys. Rev. 36, 57–64.

Sørensen, A.H., 1987. Atomic K-shell excitation at ultrarelativistic impact energies. Phys. Rev. A 36, 3125–3137.

Sørensen, A.H., Uggerhøj, E., 1986. The hunt for the density

effect in inner shell excitations. Comments At. Mol. Phys. 17, 285–309.

Spooner, D.W., Meyerhof, W.E., Kuffners, J.J., Montenegro, E.C., Ishii, K., Kuhn, S.E., Kawall, D.M., Jensen, D.G., Meziani, Z.-E., 1994. Density effect in relativistic K-shell ionisation. Z. Phys. D 29, 265–268.

Tawara, H., 1978. Cross sections for charge transfer of hydrogen beams in gases and vapors in the energy range of 10 eV to 10 keV. At. Data Nucl. Data Tables 22, 491–525.

Toburen, L.H., Nakai, M.Y., Langley, R.A., 1968. Measurement of high-energy charge-transfer cross sections for incident protons and atomic hydrogen in various gases. Phys. Rev. 171, 114–122.

Toburen, L.H., Wilson, W.E., 1977. Energy and angular distributions of electrons ejected from water vapor by 0.3–1.5 MeV. J. Chem. Phys. 66, 5202–5213.

Toburen, L.H., 1998. Ionisation and charge-transfer: basic data for track structure calculations. Radiat. Environ. Biophys. 37, 221–233.

Watanabe, N., Hayashi, H., Udagawa, Y., 1997. Bethe surface of liquid water determined by inelastic X-ray scattering spectroscopy and electron correlation effects. Bull. Chem. Soc. Jpn. 70, 719–726.

Wenzel, W.A., Whaling, W., 1952. The stopping cross section in D_2O ice. Phys. Rev. 87, 499–503.

Wilson, W.E., Miller, J.H., Toburen, L.H., Manson, S.T., 1984. Differential cross sections for ionisation of methane, ammonia and water vapor by high velocity ions. J. Chem. Phys. 80, 5631–5638.

Wilson, W.E., Nikjoo, H., 1999. A Monte Carlo code for positive ion track simulations. Radiat. Environ. Biophys. 38, 97–104.

Xu, Y.J., Khandelwal, G.S., Wilson, J.W., 1985. Proton stopping cross sections of liquid water. Phys. Rev. A 32, 629–632.

Yousif, F.B., Geddes, J., Gilbody, H.B., 1986. Balmer α emission in collisions of H, H^+, H_2^+ and H_3^+ with N_2, O_2 and H_2O. J. Phys. B 19, 217–231.

Zaider, M., Brenner, D.J., Wilson, W.E., 1983. The applications of track calculations to radiobiology I. Monte Carlo simulation of proton tracks. Radiat. Res. 95, 231–247.